2nd test

① Maxwell-Boltzmann distribution: $f(v) = 4\pi v^2 \cdot \left(\frac{m}{2\pi k_B T}\right)^{3/2} \cdot e^{-\frac{mv^2}{2k_B T}}$ $\left(m = \frac{M}{N_A}\right)$

$p = 10^5\,Pa$
$V = 10\,\ell$
$T = 300\,K$

a) Most probable speed:

$$\frac{df(v)}{dv} = 0: \quad 2v e^{-\frac{mv^2}{2k_BT}} + v^2 e^{-\frac{mv^2}{2k_BT}} \cdot \left(-\frac{m}{k_BT}\right) \cdot v = 0$$

$$v = \sqrt{\frac{2k_BT}{m}} = \sqrt{\frac{2RT}{M}} \approx 420\,\frac{m}{s}$$

b) $v_0 = 100\,m/s$

$$N_0 = N(v \le v_0) = N \cdot \int_0^{v_0} f(v)\,dv = \frac{pV}{k_BT} \int_0^{v_0} 4\pi \left(\frac{m}{2\pi k_BT}\right)^{3/2} \cdot v^2 e^{-\frac{mv^2}{2k_BT}}\,dv$$

$\frac{mv_0^2}{2k_BT} = \frac{Mv_0^2}{2RT} \approx 0{,}06 << 1$: the integrand: $v^2 e^{-\frac{mv^2}{2k_BT}} \approx v^2\left(1 - \frac{mv^2}{2k_BT}\right)$

$$N_0 = \frac{pV}{k_BT} \cdot 4\pi \left(\frac{m}{2\pi k_BT}\right)^{3/2} \left[\int_0^{v_0} v^2\,dv - \frac{m}{2k_BT} \cdot \int_0^{v_0} v^4\,dv\right] =$$

$$= \frac{pV}{k_BT} \cdot 4\pi \left(\frac{m}{2\pi k_BT}\right)^{3/2} \cdot \left(\frac{v_0^3}{3} - \frac{m}{2k_BT} \cdot \frac{v_0^5}{5}\right) = 2{,}3 \cdot 10^{21} \quad \text{(almost 1\%)}$$

② a) If the final temperature is $T_0 = 0°C$:

$$c_{Cu} m_{Cu} (T_1 - T_0) = c_i m_i^* (T_0 - T_2) + L_m \cdot m_i^* \Rightarrow m_i^* = m_{Cu} \cdot \frac{c_{Cu}(T_1 - T_0)}{c_i (T_0 - T_2) + L_m} \approx 9{,}3\,g$$

$\Rightarrow$ with 50 g ice, the final temperature does not go above 0°C.

$c_{Cu} m_{Cu} (T_1 - T_0) = 3159\,J$
$c_i m_i (T_0 - T_2) = 3157\,J$
} in the final stage ice + water + copper will be at 0°C.

$$c_{Cu} m_{Cu} (T_1 - T_0) = c_i m_i (T_0 - T_2) + L_m \cdot M \Rightarrow M = \frac{c_{Cu} m_{Cu} (T_1 - T_0) - c_i m_i (T_0 - T_2)}{L_m} =$$

$$= 8{,}5\,g$$

Entropy change:

copper: $\Delta S_{Cu} = -c_{Cu} m_{Cu} \int_{T_0}^{T_1} \frac{dT}{T} = -c_{Cu} m_{Cu} \ln \frac{T_1}{T_0} = -10\,\frac{J}{K}$

ice: $\Delta S_i = c_i m_i \int_{T_2}^{T_0} \frac{dT}{T} + \frac{M L_m}{T_0} = c_i m_i \ln \frac{T_0}{T_2} + \frac{M \cdot L_m}{T_0} \approx 12\,\frac{J}{K}$

$\Rightarrow \Delta S_{tot} = 2\,\frac{J}{K}$

b)

$T = 0°C$
$dp = 1\,atm$
$v_i - v_w = 0{,}091\,\frac{cm^3}{g}$

$$\frac{dp}{dT} = \frac{L \cdot m}{T \cdot (V_2 - V_1)} \Rightarrow dT = \frac{dp\, T (V_2 - V_1)}{L \cdot m} = -7 \cdot 10^{-3}\,K$$

(V_2: water, V_1: ice; $L = 334\,\frac{kJ}{kg}$)

(3)

Solution

(a)

$$Z_1 = \sum_{\mu} e^{-\frac{\epsilon_\mu}{k_\mathrm{B}T}} = 1 + e^{-\frac{\epsilon}{k_\mathrm{B}T}} + e^{-\frac{2\epsilon}{k_\mathrm{B}T}}$$

(b) $F = -k_\mathrm{B}T \ln Z_N = -Nk_\mathrm{B}T \ln Z_1 = -Nk_\mathrm{B}T \ln\left(1 + e^{-\frac{\epsilon}{k_\mathrm{B}T}} + e^{-\frac{2\epsilon}{k_\mathrm{B}T}}\right)$

$$S = -\left(\frac{\partial F}{\partial T}\right)_V = Nk_\mathrm{B} \ln\left(1 + e^{-\frac{\epsilon}{k_\mathrm{B}T}} + e^{-\frac{2\epsilon}{k_\mathrm{B}T}}\right) + \frac{N\epsilon}{T} \frac{e^{-\frac{\epsilon}{k_\mathrm{B}T}} + 2e^{-\frac{2\epsilon}{k_\mathrm{B}T}}}{1 + e^{-\frac{\epsilon}{k_\mathrm{B}T}} + e^{-\frac{2\epsilon}{k_\mathrm{B}T}}}$$

(c)

$$U = -\frac{\partial \ln Z}{\partial \beta} = N\epsilon \frac{e^{-\frac{\epsilon}{k_\mathrm{B}T}} + 2e^{-\frac{2\epsilon}{k_\mathrm{B}T}}}{1 + e^{-\frac{\epsilon}{k_\mathrm{B}T}} + e^{-\frac{2\epsilon}{k_\mathrm{B}T}}}$$

(d)

$$U - TS = N\epsilon \frac{e^{-\frac{\epsilon}{k_\mathrm{B}T}} + 2e^{-\frac{2\epsilon}{k_\mathrm{B}T}}}{1 + e^{-\frac{\epsilon}{k_\mathrm{B}T}} + e^{-\frac{2\epsilon}{k_\mathrm{B}T}}} - Nk_\mathrm{B}T \ln\left(1 + e^{-\frac{\epsilon}{k_\mathrm{B}T}} + e^{-\frac{2\epsilon}{k_\mathrm{B}T}}\right) - N\epsilon \frac{e^{-\frac{\epsilon}{k_\mathrm{B}T}} + 2e^{-\frac{2\epsilon}{k_\mathrm{B}T}}}{1 + e^{-\frac{\epsilon}{k_\mathrm{B}T}} + e^{-\frac{2\epsilon}{k_\mathrm{B}T}}} = F$$